

FastAPI Photo/Video Sharing

A **social media platform** for photo and video sharing, built with **FastAPI** backend, **Streamlit** frontend, and **ImageKit CDN** for media storage.

· Features

- **Media Upload** · Photo and video upload with CDN storage
 - **User Auth** · Registration and login via fastapi-users
 - **Social Feed** · Browse uploaded media from all users
 - **Async Database** · SQLAlchemy async with UUID primary keys
 - **Streamlit UI** · Clean web interface for interaction
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.. Architecture

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Streamlit Frontend · FastAPI Backend · SQLAlchemy (async) · Database
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ImageKit SDK · CDN (media storage)
```

· File Structure

```
FastAPIPhotoVideoSharing/
... main.py                # Uvicorn runner
... frontend.py            # Streamlit UI (login, upload, feed)
... app/
·   ... app.py              # FastAPI routes (auth, upload)
·   ... db.py               # SQLAlchemy async models (User, Post)
... pyproject.toml
... uv.lock
```

· Getting Started

```
cd FastAPIPhotoVideoSharing
uv sync
# Start backend
uv run python main.py
# Start frontend (separate terminal)
uv run streamlit run frontend.py
```

Environment Variables (.env)

```
IMAGEKIT_PRIVATE_KEY=...
IMAGEKIT_PUBLIC_KEY=...
IMAGEKIT_URL_ENDPOINT=...
DATABASE_URL=sqlite+aiosqlite:///./app.db
```

· Logic Flow

- 1 **Register/Login** · User authenticates via Streamlit UI
- 2 **Upload** · User selects photo/video file
- 3 **CDN Storage** · File uploaded to ImageKit, URL returned
- 4 **Post Creation** · Post record created with CDN URL and user association
- 5 **Feed** · All posts displayed in reverse-chronological order

Key Models

Model Purpose	----- -----	User UUID PK, auth via fastapi-users	Post Media URL, user FK, timestamps
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· Dependencies

FastAPI, fastapi-users, imagekitio, Streamlit, SQLAlchemy[asyncio], uvicorn

· License

Educational project · use freely for learning and reference.